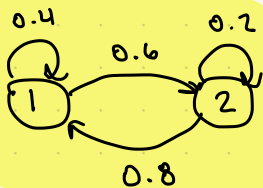


Assignment 7 Solutions

8.1



A^1 regular
has positive entries

$$A - I = \begin{bmatrix} -0.6 & 0.8 \\ 0.6 & -0.8 \end{bmatrix} \sim \begin{bmatrix} -3 & 4 \\ 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -4/3 \\ 0 & 0 \end{bmatrix}$$

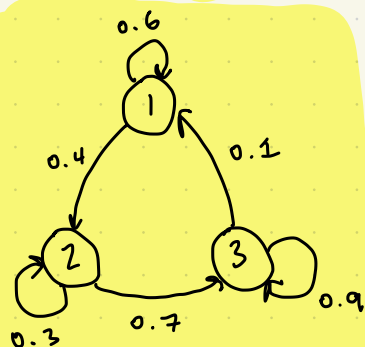
$x_1 = \frac{4}{3}x_2$
 x_2 is free

$$x_1 + x_2 = 1 \Rightarrow \frac{4}{3}x_2 + x_2 = \frac{7}{3}x_2 = 1$$

$$\text{SSV: } \begin{bmatrix} 4/7 \\ 3/7 \end{bmatrix}$$

unique

8.2



$$A^2 = \begin{bmatrix} 0.36 & 0.07 & 0.15 \\ 0.36 & 0.09 & 0.04 \\ 0.28 & 0.84 & 0.81 \end{bmatrix}$$

A^2 has positive entries
regular

$$A - I = \begin{bmatrix} -0.4 & 0 & 0.1 \\ 0.4 & -0.7 & 0 \\ 0 & 0.7 & -0.1 \end{bmatrix} \sim \begin{bmatrix} 4 & 0 & -1 \\ -4 & 7 & 0 \\ 0 & 7 & -1 \end{bmatrix} \sim \begin{bmatrix} 4 & 0 & -1 \\ 0 & 7 & -1 \\ 0 & 7 & -1 \end{bmatrix} \sim \begin{bmatrix} 4 & 0 & -1 \\ 0 & 7 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & -1/4 \\ 0 & 1 & -1/7 \\ 0 & 0 & 0 \end{bmatrix}$$

$x_1 = \frac{1}{4}x_3$
 $x_2 = \frac{1}{7}x_3$
 x_3 is free

$$x_1 + x_2 + x_3 = 1$$

$$\frac{1}{4}x_3 + \frac{1}{7}x_3 + x_3 = 1$$

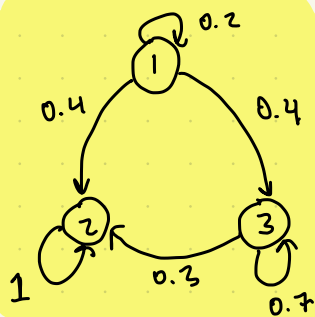
$$\frac{7+4+28}{28}x_3 = 1$$

$$\text{SSV: } \begin{bmatrix} 7/39 \\ 4/39 \\ 28/39 \end{bmatrix}$$

unique

$$\frac{39}{28}x_3 = 1 \quad x_2 = \frac{28}{39}$$

8.5



Not regular. $A^k \vec{e}_2 = \vec{e}_2$
second row of A^k

$$A - I = \begin{bmatrix} -0.8 & 0 & 0 \\ 0.4 & 0 & 0.3 \\ 0.4 & 0 & -0.3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 3 \\ 0 & 0 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x_1 = 0$$

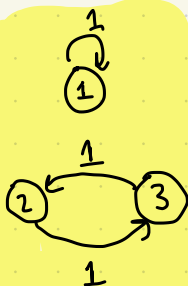
$$x_2 = 0$$

x_3 is free

SSV: $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

unique

8.6



Not Regular. $A^k \vec{e}_1 = \vec{e}_1$

$$A - I = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

x_1 is free

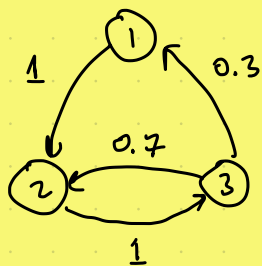
$$x_2 = x_3$$

x_3 is free

SSV: $\begin{bmatrix} 0.4 \\ 0.3 \\ 0.3 \end{bmatrix}$

not unique, e.g., $\begin{bmatrix} 0.2 \\ 0.4 \\ 0.4 \end{bmatrix}$

8.8



$$A = \begin{bmatrix} 0.21 & 0.09 & 0.147 \\ 0.49 & 0.42 & 0.433 \\ 0.3 & 0.49 & 0.42 \end{bmatrix}$$

has positive entries,

regular

$$A - I = \begin{bmatrix} -1 & 0 & 0.3 \\ 1 & -1 & 0.7 \\ 0 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -0.3 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x_1 = \frac{3}{10} x_3$$

$$x_2 = x_3$$

x_3 is free

$$x_1 + x_2 + x_3 = 1$$

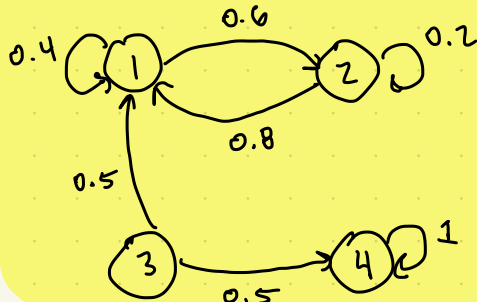
$$\frac{3}{10} x_3 + x_3 + x_3 = 1$$

$$\frac{23}{10} x_3 = 1$$

$$SSV: \begin{bmatrix} 3/23 \\ 10/23 \\ 10/23 \end{bmatrix}$$

unique

8.14

Not regular, $A^k \vec{e}_4 = \vec{e}_4$

$$A - I = \begin{bmatrix} -0.6 & 0.8 & 0.5 & 0 \\ 0.6 & -0.8 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0.5 & 0 \end{bmatrix} \sim \begin{bmatrix} -6 & 8 & 5 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 5 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -4/3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$x_1 = 4/3 x_2$
 x_2 is free
 $x_3 = 0$
 x_4 is free

$$x_1 + x_2 + x_3 + x_4 = 1$$

$$4/3 x_2 + x_2 = 1$$

$$7/3 x_2 = 1$$

$$\text{SSV: } \begin{bmatrix} 4/7 \\ 3/7 \\ 0 \\ 0 \end{bmatrix}$$

not unique

8.17 False

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$



$$\text{SSV: } \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$$

8.20 True

$$AB = A[\vec{b}_1 \dots \vec{b}_r]$$

$$= [A\vec{b}_1 \dots A\vec{b}_r]$$

$$\mathbf{1}^T (A\vec{b}_1) = \mathbf{1}^T \vec{b}_1 = 1$$

8.21 False

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

8.22 False

(see 8.5)

8.24

$$\begin{array}{c} L \\ S \\ H \end{array} \begin{array}{ccc} L & S & H \end{array} \begin{bmatrix} 0.6 & 0.1 & 0.05 \\ 0.3 & 0.7 & 0.4 \\ 0.1 & 0.2 & 0.55 \end{bmatrix}$$

$$A - I \sim \begin{bmatrix} 1 & 0 & -11/18 \\ 0 & 1 & -35/18 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x_1 = 11/18 x_3$$

$$x_2 = 35/18 x_3$$

x_3 is free

$$x_1 + x_2 + x_3 = 1$$

$$\frac{64}{18} x_3 = 1$$

$$x_3 = \frac{18}{64}$$

$$x_1 = \frac{11}{64}$$

$$x_2 = \frac{35}{64}$$

$$\vec{s} = \begin{bmatrix} 11/64 \\ 35/64 \\ 18/64 \end{bmatrix}$$

Since transition matrix is regular,

$$\lim_{k \rightarrow \infty} T^k \vec{e}_2 = \vec{s}, \text{ which indicates it's most}$$

likely to stay steady in the long term

8.27

$$a) \begin{bmatrix} p & 1-p \\ 1-p & p \end{bmatrix} = T \quad (\text{note: } T \text{ is regular})$$

$$b) T - I = \begin{bmatrix} p-1 & 1-p \\ 1-p & p-1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$$

$$x_1 = x_2$$

x_2 is free

$$\lim_{k \rightarrow \infty} T^k \vec{e}_1 = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$$